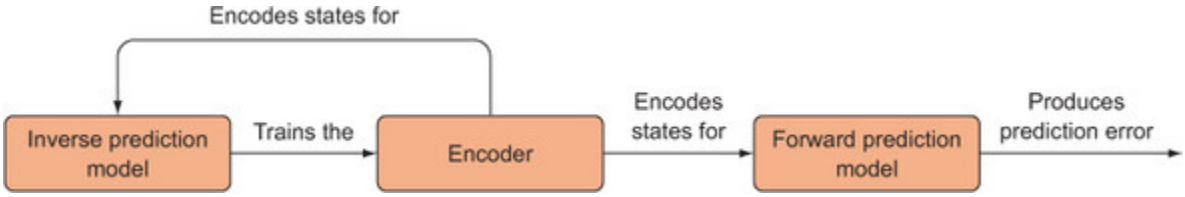
implement the intrinsic curiosity module.

8.6. Intrinsic curiosity module

As we described earlier, the intrinsic curiosity module (ICM) is composed of three independent neural network models: the forward model, inverse model, and the encoder (figure 8.13). The forward model is trained to predict the next (encoded) state, given the current (encoded) state and an action. The inverse model is trained to predict the action that was taken, given two successive (encoded) states, $\varphi(S_t)$ and $\varphi(S_{t+1})$. The encoder simply transforms a raw three-channel state into a single low-dimensional vector. The inverse model acts indirectly to train the encoder to encode states in a way that only preserves information relevant to predicting the action.

Figure 8.13. A high-level overview of the intrinsic curiosity module (ICM). The ICM has three components that are each separate neural networks. The encoder model encodes states into a low-dimensional vector, and it is trained indirectly through the inverse model, which tries to predict the action that was taken given two consecutive states. The forward model predicts the next encoded state, and its error is the prediction error that is used as the intrinsic reward.



The input and output types of each component of the ICM are shown in figure 8.14. The forward model is a simple two-layer neural network with linear layers. The input to the forward model is constructed by concatenating the state $\varphi(S_t)$ with the action a_t. The encoded state $\varphi(S_t)$ is a tensor $B \times 288$ and the action a_t : $B \times 1$ is a batch of integers indicating the action index, so we make a one-hot encoded vector by creating a vector of size 12 and setting the respective a_t index to 1. Then we concatenate these two tensors to create a batch × 288 + 12 = batch × 300 dimensional tensor. We use the rectified linear unit (ReLU) activation unit after the first layer,